

Centrifugal Pump Preventive Maintenance Manual

1. Purpose

This Standard Operating Procedure (SOP) establishes a systematic preventive maintenance program for centrifugal pumps used in an oil refinery. The purpose is to maintain reliable pump performance, reduce unplanned failures, prevent loss of containment, extend equipment life, and protect personnel and plant assets. The procedure provides routine inspection requirements, lubrication guidance, vibration and bearing-temperature monitoring, mechanical seal inspection, and troubleshooting methods for common pump faults.

2. Scope

This procedure applies to centrifugal process pumps, transfer pumps, cooling-water pumps, hydrocarbon pumps, and associated auxiliary equipment installed within refinery operating areas. It covers routine external inspection and preventive maintenance performed by trained maintenance and operations personnel. Maintenance involving opening process equipment shall be performed only after the required isolation, depressurization, draining, purging, gas testing, and permit controls have been established.

3. Roles and Responsibilities

3.1 Maintenance Technician: Perform scheduled inspections, lubrication, vibration measurements, seal checks, and corrective maintenance according to the approved maintenance plan.

3.2 Operations Personnel: Monitor pump operating condition, identify abnormal noise, leakage, vibration, temperature, pressure, or flow, and report abnormalities promptly.

3.3 Maintenance Supervisor: Verify that maintenance activities are properly planned, safely executed, documented, and completed within the required maintenance interval.

4. Maintenance Schedule

4.1 Daily Checks

1. Check pump discharge and suction pressure against normal operating values.
2. Observe for abnormal noise, vibration, pulsation, or changes in flow.
3. Inspect the mechanical seal area for visible leakage.
4. Check bearing and casing temperature for abnormal increases.
5. Verify lubrication-oil level where an oil-lubricated bearing housing is installed.
6. Inspect the pump, coupling, baseplate, and surrounding area for loose components or oil accumulation.

4.2 Weekly Checks

1. Record vibration readings at designated bearing locations.
2. Inspect coupling guards and visible coupling condition.
3. Check lubrication condition and inspect for contamination.
4. Inspect foundation bolts, mounting hardware, and accessible piping supports.
5. Compare operating readings with previous records to identify developing trends.

4.3 Monthly Checks

1. Perform a detailed visual inspection of the pump and driver.
2. Check alignment indicators and coupling condition as applicable.
3. Inspect mechanical seal leakage and gland or seal-housing condition.
4. Verify pressure gauges, temperature indicators, and other local instruments.
5. Inspect the bearing housing for oil leakage, contamination, or damaged breathers.
6. Review vibration and temperature trends for deterioration.

4.4 Quarterly Checks

1. Perform a comprehensive condition assessment of pump, motor, coupling, and foundation.
2. Verify shaft alignment using the approved alignment method when required.
3. Inspect lubrication systems and replace lubricant when required by the manufacturer's instructions or site maintenance program.
4. Inspect accessible seal components and check for abnormal wear.
5. Review maintenance history and identify recurring failures.
6. Perform additional condition-monitoring examinations when indicated by the reliability program.

5. Lubrication Requirements

1. Use only the lubricant grade specified by the pump manufacturer or approved refinery maintenance standard.
2. Maintain oil level within the marked operating range. Overfilling can cause overheating, foaming, and bearing damage.
3. For grease-lubricated bearings, apply the specified quantity at the approved interval. Avoid excessive grease application.
4. Keep grease guns, oil containers, funnels, and filling equipment clean to prevent contamination.
5. Do not mix incompatible lubricants unless compatibility has been confirmed.
6. Investigate abnormal lubricant discoloration, metallic particles, water contamination, unusual odor, or excessive consumption.
7. Replace lubricant according to manufacturer recommendations, operating severity, lubricant condition, and refinery maintenance schedule.

6. Vibration Monitoring

Vibration shall be monitored at consistent measurement locations and under comparable operating conditions. As a general screening guide, overall pump vibration around 2.8 mm/s RMS or below is commonly considered satisfactory for many industrial machines, while readings approaching 4.5 mm/s RMS require increased attention and readings around 7.1 mm/s RMS or higher should normally trigger prompt investigation. These values are guidance only; the applicable pump standard, manufacturer limits, machine class, and refinery reliability program shall take precedence.

A sudden increase from the established baseline shall be investigated even when the absolute value remains below an alarm threshold. Possible causes include misalignment, imbalance, looseness, bearing deterioration, cavitation, resonance, or hydraulic problems.

7. Mechanical Seal Inspection

1. Inspect the seal area for continuous or excessive leakage.
2. Check for crystallized deposits, staining, corrosion, or product accumulation around the seal.
3. Verify that seal flush, cooling, or barrier systems are operating at specified conditions.
4. Check accessible piping, valves, and fittings connected to the seal system.
5. Listen for unusual rubbing, squealing, or grinding sounds.
6. Monitor seal temperature when instrumentation is provided.
7. If leakage becomes excessive, product release occurs, or seal performance deteriorates rapidly, stop or isolate the pump according to approved operating procedures and notify maintenance.

8. Bearing Temperature Limits

Bearing temperature shall be monitored against the manufacturer's specified operating limit. As a general maintenance guideline, a sustained bearing temperature above approximately 80°C should be investigated, while temperatures approaching 90°C or a rapid temperature increase should normally be treated as an urgent abnormal condition. Manufacturer limits and refinery engineering standards take precedence.

Possible causes include inadequate or excessive lubrication, contaminated lubricant, bearing wear, shaft misalignment, excessive hydraulic load, or cooling problems.

9. Troubleshooting Table

Fault	Possible Causes	Corrective Action
Fault	Possible Causes	Corrective Action
Cavitation	Low suction pressure, restricted suction line, inadequate NPSH, excessive liquid temperature	Check suction pressure and valve position; inspect suction strainer and piping; verify operating conditions and NPSH requirements.
Leakage	Damaged mechanical seal, worn shaft sleeve, loose connection, excessive vibration	Inspect seal and connections; check alignment and vibration; replace damaged components after safe isolation.
Overheating	Poor lubrication, excessive grease, bearing damage, misalignment, excessive load	Check lubricant level and condition; inspect bearings; verify alignment and operating load.
High vibration	Misalignment, imbalance, looseness, bearing wear, cavitation	Check alignment, coupling, foundation, bearings, and hydraulic conditions; compare with vibration trends.
Low flow	Blocked suction/discharge, incorrect valve position, impeller damage, low speed	Check valves and piping; verify speed and pressure; inspect impeller when safely isolated.

Fault	Possible Causes	Corrective Action
Unusual noise	Cavitation, bearing damage, rubbing, foreign material	Check suction conditions, bearing condition, clearances, and pump internals.

10. Safety Requirements

Before intrusive maintenance, isolate the pump from all energy and process sources. Apply lockout/tagout as required, depressurize and drain equipment, verify zero hazardous energy, and perform gas testing where required. Use appropriate PPE and follow refinery permit-to-work and confined-space requirements.

11. Documentation

Record inspection dates, operating conditions, vibration readings, bearing temperatures, lubricant condition, seal observations, defects, corrective actions, and replacement components. Trend vibration and temperature data to identify deterioration before failure. Maintenance records shall be retained according to refinery equipment-management and document-control requirements.

12. Completion

After maintenance, verify that all tools, temporary equipment, and protective materials have been removed. Confirm guards are installed, valves and auxiliary systems are correctly positioned, and the pump is ready for controlled return to service. Conduct a post-maintenance inspection and record the equipment's operating condition.